

Language Models as Reads over Atom Spaces: A Unified Measure-Theoretic Formulation of Context, Retrieval, and Parameters

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1 **Research Highlights**

- 2 • Weighted neural reads are expressed over valued measurable atom spaces.
- 3 • Exact top- k read truncation has two exponential error bounds.
- 4 • CounterFact-500 exposes strong retrieval but weak thresholded paraphrase
5 recall.
- 6 • Exact tail mass selects k under a certified vector-error budget.
- 7 • Signed-read perturbation bounds quantify approximate cancellation failure.
- 8 • Five-seed matched routing exposes corpus-to-parameter confusion.

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